One Policy per Sequence, Selected Without Ground Truth

Extended Abstract

Luai Abuelsamen[1]
[1]University of California, Berkeley, luai_abuelsamen@berkeley.edu

Abstract: We enter all three tiers of Track 2 with one FlashCHORD policy per sequence on a Unitree G1 with Dex3 hands. The tiers share a ground truth and differ only in the *input* a submission may condition on. Everything per-clip that we ship reads the input alone: a registered selection rule reproducible without ground truth, and, on Tier 1, training and warm-start fine-tuning against the cleaner reference that tier permits. The latter is our largest measured improvement, at $0.57 per clip; substituting the same reference under an already-trained policy instead reduces AUC by 0.18.

1. Introduction

Track 2 asks for physically realisable object trajectories for 30 manipulation sequences, produced by a policy rollout rather than by reconstruction, at three tiers differing in the reference a submission may condition on: a clean multi-view trajectory (Tier 1), a noisy released retarget (Tier 2), and whatever the entrant reconstructs from video (Tier 3).

We found our first result by accident: our Tier 1 and Tier 2 public scores came back *identical*. The uploaded files are byte-identical, and a Kabsch fit shows every shipped rollout's reference to be an exact (0.000 cm) rigid transform of the released Tier 2 trajectory. All three tiers are scored against the same ground truth; only the permitted input differs.

We therefore focused on methods that use only the reference each tier permits. Unlike physics-based human–object imitation [4, 6], which assumes a clean motion-capture reference, here the reference is itself an estimate.

Contributions:

- a per-episode selection rule reading only the *input* reference: legal in every tier, reproducible without ground truth;
- a measurement separating two uses of a cleaner reference: substituted under a trained policy it costs 0.18 AUC, while a warm-start fine-tune to it gains 0.08.

2. Approach

Pipeline. SOMA-X parameters are retargeted to the G1 with Dex3 hands on CPU (30/30 clips, 0.0 mm residual). FlashCHORD SAC trains one policy per sequence for 349,999,104 steps at `world_count=1` with `scene.support` set per clip from the shipped primitives, then the challenge evaluator, selection and the kit's packer.

Recipe changes. Four ship: an object-*orientation* reward term (weight 1.0, std 0.3), which the configuration lacked despite having a position weight; a start-curriculum patch; object-position std tightened $0.2 \rightarrow 0.08$ m; and finger targets on bridge-smoothed references. Each refuses to apply to an unrecognised upstream, so a half-patched run cannot reach the GPU.

Selection, registered before the run. The provenance schema permits a different `checkpoint_sha256` per episode, so we pick, per clip, the latest checkpoint whose achieved object never leaves a 10 cm band around the *input* reference. On Tier 2 this is a cost: 13 rollouts kit-score higher than the ones that ship, and we shipped the rule's picks.

Tier 3. We run the challenge's own modules in order: monocular geometry [3], open-set detection [1], segmentation [2] and 6D tracking [5]. Placement in the released world is a single rigid transform taken from frame 0, as the FAQ permits, with no scale fitted and no per-frame correction.

Warm start. A new configuration key loads a shipped actor's parameters and batch statistics after learner construction and advances the step counter so the curriculum resumes in its final stage, then runs to the normal horizon: 50M steps, 35 minutes on an A6000. The critic starts cold, so the actor's learning rate is scaled $\times 0.1$ for 2,000 updates.

Entry	*AUC*	*SP-SR*	*MP-SR*	*MPPE*
Tier 1 (ours)	**0.605**	0.733	0.633	**11.72**
CHORD base.	0.510	—	—	—
Tier 2 (ours)	**0.465**	0.667	0.467	**14.13**
CHORD base.	0.374	0.533	0.067	20.50
Tier 3 (ours)	0.268	0.200	0.033	47.77

Table 1: All 30 clips, one seed, official kit scorer; MPPE in cm, lower better.

Scoring. We did not reimplement the metrics; our scorer hash-checks the kit's metric files against the shipped manifest and calls them. Across nineteen uploads the largest local-to-public AUC disagreement was 1.1×10^{-5}.

3. Experiments and Results

The largest single effect was a bug. Our first full-length baseline scored 0.043 mean AUC because every run script passed `scene.support=false`: without the table the object falls at reset and the episode ends on frame 0.

Three ways to use a cleaner reference. Tier 1 permits the clean multi-view trajectory, and how a policy is given it decides whether it helps. Substituted under the 30 shipped actors at evaluation time, mean AUC *fell* from 0.465 to 0.289 and SP-SR from 0.667 to 0.138, with only 2 of 30 clips improving; frame checks were below 1.7×10^{-5} cm, so this is distribution shift rather than a conversion artefact — each actor had learned one trajectory. Warm-start fine-tuning to it, under a no-regression adoption rule, instead took Tier 1 to 0.548 AUC at 12.06 cm. The gain comes from re-fitting rather than extra training: a second generation on the *same* reference adopted 0 of 3. Whether it works is predicted by a clip's shipped score: 11 of 19 clips above 0.35 adopted, against 0 of 4 below it. Training from scratch on the clean reference was tried last, on the seven clips nearest the 3 cm MP-SR threshold: 6 of 7 adopted, gaining +0.11 to +0.53 AUC each and taking the entry to the 0.605 at 11.72 cm in Table 1. On Tier 3, where selection may read only our own reconstruction, 5 of 10 adopted, taking that entry from 0.232 to 0.251; a later batch of fine-tunes took it to the 0.268 reported in Table 1.

Effect of checkpoint selection. The value of selection depends on the run. On the 30-clip Tier 2 run the registered rule is worth +0.0010 AUC over the final checkpoint, and an oracle reading ground truth over that run's own checkpoints reaches 0.420. On the seven Tier 1 clips trained from scratch it is worth +0.057, because those curves are not monotone: one clip peaks at 300M steps and falls to 0.387 AUC by 350M while two others peak at 350M. There the rule chose the same checkpoint as a ground-truth-reading selection on 6 of 6 adopted clips. The standard deviation of AUC over a clip's last five checkpoints averages 0.081 — checkpoint-to-checkpoint variability within a run, not a between-seed estimate, but larger than several effects we measured.

Where the gap is. Measured on trajectories rather than on AUC, the noisy Tier 2 reference sits 1.94 cm from truth while our policies sit 13.84 cm, and *19 of 30 clips end up worse than the reference they were given.* Only 9 rollouts ever leave a 10 cm band around it; the rest track their reference within a few centimetres. The main deficit is therefore persistent trajectory error across many clips, alongside larger failures in a smaller subset.

4. Conclusion

Both results above concern how a reference is supplied to a policy rather than how the policy is rewarded, and neither is specific to this benchmark. The residual failure is mid-episode release of a well-tracked object, and one second before each loss nothing observable differs from a healthy frame. Inspection of the environment implementation shows that a per-link contact state is computed every step and feeds a reward term carrying the largest positive weight in the objective, but is not included in the actor's observation. This may limit the policy's ability to react to a changing contact, and it is consistent with what the contact-conditioned literature reports [4, 6], but we did not test the explanation experimentally. A contact-conditioned ablation is the first thing we would run next.

Cost and reproducibility. Spend was $166 for 30 policies across three tiers, at $3.00 per full-length training run and $0.57 per warm-start fine-tune. Selection and adoption read only the permitted input, so rollouts are judged without ground-truth annotation. The pipeline is one CLI whose configuration patches refuse to apply to an unrecognised upstream, and whose scorer calls the kit's own metric code under a hash check.

Limitations. A single seed, and no ablation isolates the four recipe changes. Object inertia is inherited unchanged from the simulator, so every object is equally easy to rotate.

References

[1] Shilong Liu et al. Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection. *arXiv:2303.05499*, 2023.

[2] Nikhila Ravi et al. SAM 2: Segment anything in images and videos. *arXiv:2408.00714*, 2024.

[3] Ruicheng Wang et al. MoGe: Unlocking accurate monocular geometry estimation for open-domain images. *arXiv:2410.19115*, 2024.

[4] Yinhuai Wang et al. PhysHOI: Physics-based imitation of dynamic human-object interaction. *arXiv:2312.04393*, 2023.

[5] Bowen Wen et al. FoundationPose: Unified 6D pose estimation and tracking of novel objects. In *CVPR*, 2024.

[6] Sirui Xu et al. InterMimic: Towards universal whole-body control for physics-based human-object interactions. In *CVPR*, 2025.